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LoReCast: A LongFormer-Based Framework with Reinforcement Learning for News Article Summarization

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ABSTRACT

This study introduces LoReCast, a LongFormer-based framework integrated with Reinforcement Learning (RL), designed to generate concise, coherent, and contextually rich summaries of news articles. The system aims to improve information accessibility by reducing redundancy and preserving key details in long-form content. The framework leverages LongFormer's sparse attention mechanism to handle lengthy transcripts efficiently, while Reinforcement Learning fine-tunes the summarization process through a multi-objective reward function that considers ROUGE, BERTScore, Repetition Rate, and Flesch Reading Ease. Results demonstrate that LoReCast outperforms baseline summarization models by producing summaries that are more readable, contextually coherent, and semantically aligned with human-written references. This study shows that adding reinforcement learning to transformer models can make text summarization faster and more accurate for journalistic articles.

Keywords— LongFormer, Reinforcement Learning, News Article, Automatic Text Summarization, Exposure Bias, Maximum Likelihood Estimation

I. INTRODUCTION

The exponential growth of digital information has made it increasingly difficult for users to identify and extract relevant content from a vast array of sources. Studies indicate that younger individuals and low-income populations are significantly less likely to engage with traditional news due to time constraints, digital access gaps, and a perceived lack of trust or relevance in available formats [1]. This decline in active news consumption underscores a critical need for automated summarization systems that can bridge the information gap. By selecting and condensing the most informative elements from one or multiple documents, these systems serve as efficient alternatives to reading entire texts, significantly reducing the time required to access vital information across various domains, including news, lectures, and multi-party meetings.

Despite the evolution of automatic text summarization from text-based approaches in the 1950s to modern neural models, significant challenges remain [2]. Traditional models often rely on Maximum Likelihood Estimation (MLE) for training, which is limited by exposure bias. This phenomenon occurs because the model is trained to predict the next token based on ground-truth context (teacher forcing) but must rely on its own generated outputs during inference, leading to error accumulation and high repetition rates [3]. Furthermore, popular encoder models like BERT are restricted by a 512-token input limit, rendering them unsuitable for long-form content. While the introduction of the LongFormer architecture—utilizing a sparse attention mechanism—addresses this by extending input capacity to thousands of tokens with linear computational complexity, persistent issues regarding redundancy and structural disjointedness continue to compromise the conciseness and readability of the generated outputs [4].

To address these limitations, this study proposes LoReCast: a LongFormer-based summarization framework optimized through Reinforcement Learning (RL). By integrating RL into the architecture, the system moves beyond word-level predictions to a sequence-level perspective, employing a reward-driven optimization strategy that penalizes redundancy and improves summary quality through dynamic policy updates. Unlike early on-policy RL methods that suffered from instability and high computational costs [5], this approach balances stability and exploration to prioritize relevance and fluency. This allows the model to generate abstractive summaries that are more coherent and concise than those produced by traditional extractive methods, which often struggle with repetitive content in written news articles.

Furthermore, the LoReCast system enhances user accessibility by incorporating a Zero-Shot Classification module. Utilizing a large pre-trained language model, such as BART-MNLI, the framework automatically categorizes summaries into predefined topic labels—including politics, health, or disaster recovery—without requiring labeled training data. This semantic alignment allows for better organization and searchability, making the system highly adaptive for users who prefer content filtered by thematic relevance. By combining the extended input capacity of LongFormer with the optimization power of RL and the interpretability of zero-shot tagging, this research offers a robust solution to the persistent challenges of repetition and fragmentation in long-form news summarization.

II. EXPERIMENTAL METHOD/S

A. Research Design

This study's research design adopts an experimental approach to evaluate an abstractive summarization system for news articles, integrating LongFormer with Reinforcement Learning (RL) to address issues of readability and subjective quality. Building upon the foundational framework outlined in "A LongFormer-Based Framework for Accurate and Efficient Medical Text Summarization," Our research advances the architecture by incorporating RL-based optimization mechanisms tailored to improve output quality and reduce repetitive content. The study employs a multi-stage design wherein the LongFormer encoder processes lengthy input texts using sparse attention to manage token complexity efficiently. A sequence-to-sequence (Seq2Seq) decoder generates initial summaries, which are then refined through a custom RL module. This reinforcement learning component utilizes reward functions based on ROUGE scores, redundancy penalties, and readability metrics to dynamically adjust the model's content selection and phrasing strategies. Key to this experimental design is the comparative analysis between two models: the baseline LongFormer-only summarizer and the LongFormer-

based framework with an RL system. Using performance metrics such as ROUGE scores, BERTscore, repetition rate, and the Flesch Reading Ease, the study systematically measures the impact of RL integration on summary quality. The use of the CNN/DailyMail dataset for both pretraining and evaluation ensures the robustness and real-world applicability of the summarization model's performance. This design emphasizes both quantitative evaluation and performance benchmarking, aiming to provide measurable improvements in conciseness, accuracy, and interpretability of long-form news article content. The research not only tackles the practical challenge of enhancing user comprehension of complex documents but also contributes to the broader field of Natural Language Processing by refining transformer-based summarization through reinforcement learning.

B. Sources of Data

The study uses the pre-existing CNN/DailyMail dataset, publicly available on Kaggle, as the primary dataset for training and evaluating the proposed LongFormer-based summarization framework. This benchmark dataset contains thousands of long-form news articles paired with professionally written highlights, serving as reference summaries for supervised learning. The diversity of topics and language styles in the CNN/DailyMail corpus provides a robust foundation for training models to recognize salient content and generate coherent summaries. The dataset is useful for both the initial training of the LongFormer encoder-decoder architecture and the reinforcement learning phase, where reward functions are applied to fine-tune summary quality. Each article-summary pair supports the development of content selection and abstraction strategies, with reinforcement learning enhancing these outputs by penalizing redundancy and encouraging clarity and conciseness. By using this well-established dataset, the study ensures empirical rigor and enables meaningful comparisons with existing summarization models. The CNN/DailyMail corpus forms the experimental backbone of the research, offering a scalable and linguistically diverse collection that supports the systematic evaluation of the model's ability to manage real-world, information-rich news article content.

C. Research Instrument

The research instrumentation follows a structured five-phase experimental design to validate the LoReCast framework. Experiment Paper 1 (Data Generation) establishes the foundational dataset by extracting long-form news articles and human-written references from the CNN/DailyMail corpus to ensure data consistency. Experiment Paper 2 (Lexical Evaluation) utilizes ROUGE metrics to measure word-level accuracy and n-gram overlap, while Experiment Paper 3 (Semantic Evaluation) employs BERTScores to assess how well the system preserves the core meaning through contextual embeddings. To provide a holistic view of the system's utility, Experiment Paper 4 (Comprehensive Performance Evaluation) aggregates these results across a 10,000-article test set while incorporating Repetition Rate and Flesch Reading Ease to measure conciseness and readability. Finally, Experiment Paper 5 (Statistical Analysis) applies a paired t-test to these metrics to determine if the performance improvements of the LoReCast model over the baseline are statistically significant.

D. Software and Hardware Tools

The researchers selected Visual Studio Code due to its lightweight design, ease of use, and extensive support for extensions tailored to deep learning development. It offers syntax highlighting, integrated terminals, Git integration, and debugger support, making it an efficient IDE for managing codebases in natural language processing tasks.

Python was chosen as the programming language because of its robust ecosystem, which includes a wide array of libraries for NLP, deep learning, and reinforcement learning. Its versatility and community support enable rapid prototyping and flexible integration of machine learning components.

The Hugging Face Transformers library was used to integrate the LongFormer model, which forms the core of the summarization framework. LongFormer introduces a sparse attention mechanism that enables processing of long documents (up to 4096 tokens) while maintaining computational efficiency. This is critical in summarizing lengthy and information-rich news articles. The library also supports pre-trained weights and tokenizers, simplifying model loading and customization.

The study utilizes PyTorch as the primary deep learning framework to construct the summarization model and implement the REINFORCE algorithm, a policy gradient method in reinforcement learning. A custom REINFORCE module is developed to fine-tune the model by guiding it toward generating high-quality summaries. This module operates by first generating candidate summaries based on the model's current policy, then evaluating these outputs using a defined reward function. The reward function integrates multiple quality metrics, including ROUGE scores, redundancy penalties, and readability assessments, to quantify the effectiveness of each summary. Based on the resulting reward signals, gradient updates are applied to optimize the model's parameters, encouraging the production of concise, coherent, and informative summaries in future iterations.

To ensure the reliability and validity of the generated summaries, the study employs a combination of automated evaluation metrics and supporting libraries to assess the performance of the summarization model. For lexical overlap, the ROUGE evaluation toolkit (py-rouge or rouge-score) is used to compute ROUGE scores, providing insights into informativeness and structure. To measure semantic similarity, the study uses the bert-score Python library, which leverages pre-trained BERT models from Hugging Face Transformers to calculate precision, recall, and F1 scores based on contextual embeddings. Redundancy is evaluated using custom Python functions written in NumPy and NLTK, which compute the Repetition Rate by comparing unique versus total tokens in each summary. Lastly, Flesch Reading Ease is assessed using the textstat Python library, which computes readability scores, including FRE based on sentence length and syllable complexity. These tools collectively enable a comprehensive and objective analysis of the model-generated summaries in terms of relevance, semantic fidelity, conciseness, and accessibility.

E. System Architecture

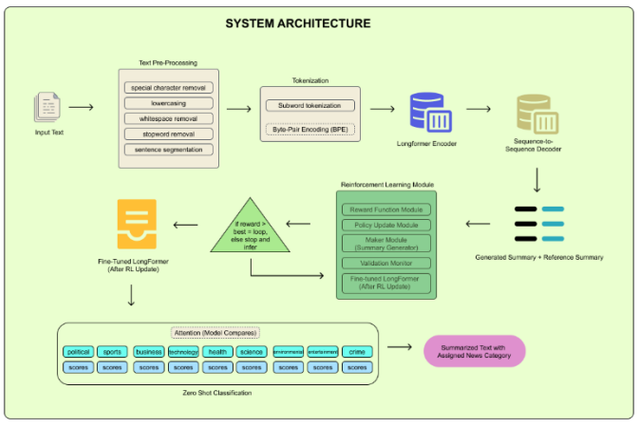


Figure 1. Longformer-Based Framework With Reinforcement Learning and Zero Shot Classification

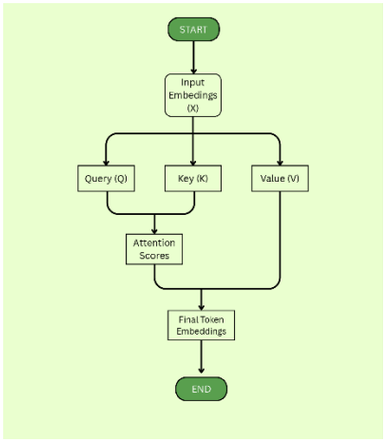


Figure 2. Process of Longformer-Encoder

This is applied to the processed text and LoreCast model outputs to handle long inputs and generate a summary in a sequence-to-sequence manner. The LongFormer encoder is under a supervised maximum-likelihood estimation (MLE) regime where the encoder reads the article's token sequence through a sparse-attention mechanism that efficiently handles long inputs. This allows up to 4096 tokens to be handled.

Input Text

The process begins with the acquisition of long-form news articles from the CNN/DailyMail dataset, which contains over 300,000 unique documents. For this study, a sample of 10,000 unique articles is selected to ensure a diverse and manageable dataset. These articles serve as the primary source material, each paired with a human-written reference summary that acts as the gold standard for evaluation.

Text Processing and Tokenization

Before summarization, the input undergoes a multi-stage preprocessing phase to ensure data consistency. This involves cleaning unnecessary symbols, sentence segmentation, and stopword removal. To further normalize the text, the system applies lemmatization, stemming, and Part-of-Speech (POS) tagging to capture the syntactic relationships between words. For tokenization, LoReCast utilizes byte-pair encoding (BPE). Unlike traditional word-based splitting, BPE handles rare terms by breaking them into subword units, which optimizes computational efficiency while preserving the semantic meaning of lengthy news transcripts.

LongFormer Encoder and Seq2Seq Decoder

The core architecture applies the processed tokens to a LongFormer encoder-decoder model. The encoder operates under a supervised maximum-likelihood estimation (MLE) regime, using a sparse-attention mechanism to handle inputs of up to 4096 tokens. The resulting

representations are passed to a Sequence-to-Sequence (Seq2Seq) decoder, which is trained using a teacher-forcing approach. During this phase, the decoder predicts the next token based on the ground-truth reference from the previous position. The system uses cross-entropy loss and the AdamW optimizer with gradient clipping to update the weights of both the encoder and decoder.

Reinforcement Learning Module

To refine the initial outputs, the framework incorporates a reinforcement learning (RL) module that optimizes the model using a composite reward function. This reward, R, integrates ROUGE scores for lexical accuracy, a redundancy penalty () that reduces the repetition of n-grams, and a fairness coverage score () to ensure key topics from the reference summary are included. The model employs the REINFORCE policy-gradient algorithm, where an advantage (A) is calculated by comparing the current reward to a baseline (b). This feedback loop allows the model to minimize the reinforcement learning loss (L_{RL}), fine-tuning the parameters to produce summaries that are more coherent and less repetitive than those produced by standard MLE training alone.

Zero-Shot Classification

Following the generation of the summarized text, it is passed into a Zero-Shot Classification module to enhance organization and searchability. This module utilizes the *bart-large-mnli* model to categorize the summary into predefined labels such as politics, health, or economy. By framing the task as a natural language inference (NLI) problem, the system evaluates the semantic alignment between the summary and candidate hypotheses without requiring labeled training data for each specific category. The label with the highest confidence score is assigned, resulting in a finalized, categorized summary output.

F. Data Gathering and Data Generation Procedure

The study utilizes the CNN/DailyMail dataset, a benchmark corpus consisting of long-form news articles paired with human-written summaries. A total of 50,000 articles were extracted and partitioned using a 60:20:20 split, resulting in 30,000 articles for supervised training, 10,000 for validation and hyperparameter tuning, and 10,000 for final performance testing on unseen data.

To ensure consistency, the dataset undergoes a preprocessing pipeline that includes lowercasing, the removal of numerical characters and punctuation, and sentence segmentation. The refined text is then processed using a LongFormer-compatible tokenizer to generate subword tokens, numerical IDs, and the specific attention masks required for the model's sparse attention mechanism. This structured approach prepares the data for efficient long-range encoding and subsequent optimization.

H. Ethical Considerations

This study does not involve direct collection of data from individuals and does not process any personally identifiable information. As such, it does not require informed consent or raise concerns related to human subject research. Ethical standards were upheld by ensuring that the use of third-party data, such as the CNN/DailyMail dataset, complied with its respective usage license and citation requirements. The research process maintained academic integrity by avoiding data manipulation or misrepresentation of model performance. All results were reported transparently, and fairness in system evaluation was prioritized by using standardized and reproducible metrics such as ROUGE, BERTScore, Repetition Rate, and Flesch Reading Ease. Furthermore, any biases or limitations of the model were documented to promote accountability and responsible AI development.

III. RESULTS AND DISCUSSION

The baseline performance of the LongFormer framework was evaluated using a test sample of 10,000 articles from the CNN/DailyMail dataset to establish technical benchmarks before Reinforcement Learning optimization. The recorded metrics—specifically a ROUGE-1 of 0.2765, ROUGE-2 of 0.0464, and ROUGE-L of 0.1782—indicate that the model performs below the excellence benchmarks for unigram coverage and bigram fluency. This discrepancy is primarily attributed to the model's abstractive nature, where the generation of novel phrasing creates a "lexical gap" that exact-match metrics cannot accommodate [6]. While synonym usage and reordering suppress ROUGE values, the baseline achieved a rescaled BERTScore of 0.2349, falling within the "Good/Fair" interpretability range. This suggests that while the framework may fail exact-match requirements, it maintains underlying semantic groundedness through "meaning-match" success [6].

The most significant computational challenge identified in the baseline model is the 38.01% repetition rate, which signifies a state of "textual degeneration." This high degree of redundancy is a technical failure rooted in Maximum Likelihood Estimation (MLE) training and the resulting exposure bias. Because the model is exposed only to ground-truth human data during training, it fails to learn self-correction for suboptimal predictions during independent generation. In long-sequence tasks, a single repetitive bigram often triggers a self-reinforcing feedback loop, where recently generated words increase the probability of their own re-selection. Consequently, nearly 40% of the token budget is wasted on redundant phrases, leading to a "semantic collapse" that prevents the system from capturing a diverse range of facts from the source text [7].

Furthermore, the baseline's Flesch Reading Ease score of 74.00, while categorized as "Fairly Easy," is interpreted as a superficial byproduct of simplistic and repetitive syntactic structures. Rather than achieving true clarity, the model defaults to a baseline simplicity comparable to a 7th-grade reading level, lacking the nuanced communication necessary for professional news abstraction

[8]. These empirical findings demonstrate that human-written highlights, which contain virtually zero n-gram repetition, remain far superior to the baseline output. This data provides the primary justification for transitioning to a Reinforcement Learning framework, where a reward-based system can explicitly penalize redundancy and force the model to prioritize global information diversity and narrative flow.

The evaluation of the LoReCast framework demonstrates the model's capacity to transcend the technical limitations observed in the baseline LongFormer. As illustrated by the comparison with the results in Table I, the integration of Reinforcement Learning (RL) triggered a substantial increase across all lexical and semantic metrics [5]. The ROUGE-1 score improved to 0.3573, indicating that the RL agent effectively incentivized the model to identify and include a higher volume of critical unigrams from the source text. Most notably, the ROUGE-2 score saw a nearly threefold increase to 0.1189. This improvement confirms that by optimizing for discrete linguistic rewards rather than simple token probability, the model significantly enhanced its bigram fluency and the preservation of logical word-pair relationships that are often lost in standard training.

The most definitive proof of the framework's success is the recorded BERTScore of 0.3210. Following the interpretability scales established for transformer models, scores exceeding 0.30 are categorized as "Excellent," marking a significant advancement from the baseline's 0.2349. This achievement confirms that LoReCast has moved beyond surface-level word matching into a state of high-level semantic alignment. Unlike the baseline, the RL-optimized model demonstrates a sophisticated understanding of the source material's meaning, ensuring that the generated summaries are not only lexically dense but also semantically faithful to the human gold standard. This shift confirms that reward-based optimization allows the model to prioritize global context over local word patterns [6].

A critical success identified in this phase is the reduction of the repetition rate to 27.47%, a 10.54% decrease from the baseline's 38.01%. This reduction directly addresses the computational problem of exposure bias by incorporating a reward function that explicitly penalizes redundancy. By breaking the self-reinforcing loops that previously led to textual degeneration, the framework successfully forces the model to prioritize information diversity over the repetitive, high-probability local choices typical of traditional training. While the rate remains higher than human linguistic patterns, this significant drop proves that RL effectively mitigates the "looping" behavior that compromises the baseline's utility [5] [7].

Finally, the Flesch Reading Ease score shifted from 74.00 to 60.84, moving from the "Fairly Easy" category to "Standard/Plain English." According to established readability thresholds, a score of 60–70 represents the ideal level for a general audience [8]. In the context of this study, this decrease in the "ease" score signifies a transition toward more sophisticated and human-like sentence structures. While the higher baseline score was a superficial result of simplistic and repetitive syntax, the lower score for LoReCast reflects the generation of more complex and informative sentences. Collectively, these results demonstrate that the transition to a Reinforcement Learning framework successfully bridges the performance gap, resulting in abstractive summaries that are semantically grounded and structurally coherent.

The statistical significance of the performance gains achieved by the LoReCast framework was evaluated through a paired t-test conducted over the 10,000-article test set. As detailed in the statistical analysis, the results provide overwhelming empirical evidence that the integration of Reinforcement Learning (RL) produces a superior approach to abstractive summarization. The mean differences for ROUGE-1 (+0.0808), ROUGE-2 (+0.0725), and ROUGE-L (+0.0479) all returned a p-value of 0.0000 ($p < 0.001$), confirming that the observed improvements in content coverage and structural coherence are not the result of random variation. The high t-values, particularly the 116.245 recorded for ROUGE-1, demonstrate a massive effect size across the large-scale dataset, while the BERTScore F1 increase of +0.0861 further validates that the framework significantly enhances semantic alignment.

A critical finding in this statistical evaluation is the significant reduction in the Repetition Rate, which saw a mean difference of -10.5537. The robust negative t-value of -128.366 confirms that the RL agent's reward function successfully penalized the "textual degeneration" loops that compromised the baseline model. By explicitly targeting the redundancy problem, LoReCast achieved a more varied and informative output, effectively mitigating the exposure bias that previously caused the system to exhaust its generation budget on repetitive phrases. This is particularly vital for maintaining the integrity of short articles where the baseline often fails to provide a comprehensive overview before entering a repetition loop.

The analysis also revealed a significant decrease in the Flesch Reading Ease score of -13.168 ($t = -114.357$). While a lower score typically suggests reduced readability, in the context of neural text generation, this shift represents an improvement in linguistic quality. The baseline's higher score was a byproduct of simplistic and repetitive syntactic structures, whereas the lower score for LoReCast indicates the generation of more complex and varied sentences. By moving into the "Standard/Plain English" range, the model produces summaries that reflect the professional standards of news journalism [8]. The consistent p-values of 0.0000 across all metrics provide the highest level of statistical confidence that the transition to Reinforcement Learning creates summaries that are more accurate, less redundant, and more semantically
aligned with human gold standards.

IV. CONCLUSION

The results of this study establish that textual redundancy is the most significant computational hurdle in baseline long-form summarization. The high repetition rate of 38.01% observed in the LongFormer baseline confirms the presence of textual degeneration, a direct consequence of exposure bias inherent in traditional Maximum Likelihood Estimation (MLE) training. By integrating Reinforcement Learning (RL), the LoReCast framework successfully mitigated this issue, reducing the repetition rate to 27.47% ($p < 0.001$). This reduction allowed the system to more effectively allocate its generation budget toward diverse and relevant content, leading to a substantial improvement in lexical density and content coverage. Furthermore, the shift in the Flesch Reading Ease score from 74.00 to 60.84 signifies an advancement toward more sophisticated, human-like syntactic structures, moving beyond the simplistic and repetitive outputs typical of MLE-trained models.

The study also highlights the critical distinction between lexical overlap and semantic integrity in abstractive summarization. While the LoReCast framework significantly outperformed the baseline, neither model reached the "Excellence" thresholds for ROUGE scores

(R1 > 0.40; R2 > 0.20) due to their abstractive nature. The "lexical gap" created by the generation of novel phrasing and synonyms naturally suppresses exact-match metrics. However, the achievement of an "Excellent" BERTScore of 0.3210 proves that the model successfully captures semantic intent and global meaning, even when literal n-gram overlap is constrained. These findings demonstrate that semantic-aware metrics provide a more accurate reflection of abstractive quality than traditional word-matching techniques.

In summary, the paired t-test results across all performance metrics ($p < 0.001$) validate that the transition from a purely supervised training regime to a reward-driven Reinforcement Learning framework provides a robust solution to the limitations of exposure bias. By incorporating feedback mechanisms that explicitly penalize redundancy and reward information diversity, the LoReCast framework ensures the production of summaries that are semantically grounded, structurally coherent, and linguistically complex. These results provide a strong empirical foundation for the continued use of reinforcement learning to bridge the performance gap between machine-generated abstractions and human linguistic standards in news summarization.

V. REFERENCES

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*Appendices/Supplementary Data

Table 1. SOP 1: The summarization performance of the LongFormer base framework without Reinforcement Learning

Metric	**Value**
ROUGE-1	0.2765
ROUGE-2	0.464
ROUGE-L	0.1782
BertScore	0.2349
Repetition Rate	38.01%
Flesch Reading Ease	74.00

Table 2. SOP 2: The summarization performance of the LongFormer base framework with Reinforcement Learning

Metric	Value
ROUGE-1	0.3573
ROUGE-2	0.1189
ROUGE-L	0.2261
BertScore	0.3210
Repetition Rate	27.47%
Flesch Reading Ease	60.84

Table 3. SOP 3: Significant differences between the Longformer-based framework with Reinforcement Learning and without Reinforcement Learning in summarizing news articles

Metric	Mean Difference	t-value	p-value	Significance	Interpretation
ROUGE-1	+0.0808	116.245	0.0000	Significant	The study is significantly better (higher mean)
ROUGE-2	+0.0725	106.628	0.0000	Significant	The study is significantly better (higher mean)
ROUGE-L	+0.0479	79.047	0.0000	Significant	The study is significantly better (higher mean)
Bert Score	+0.0861	92.346	0.0000	Significant	The study is significantly better (higher mean)
Repetition Rate	-10.5537	-128.366	0.0000	Significant	The study is significantly better (lower mean)
Flesch Reading Ease	-13.168	-114.357	0.0000	Significant	The study is significantly worse (lower mean)

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